TECHNICAL OVERVIEW

9 October 2026

Contracted Growth

Classifiers that absorb new classes without changing the answers they have already given

A deployed classifier that must learn new categories today has two options. Retrain it, which costs a full training run and requires that all the original data still be held; or update its weights on the new data, which is cheap but silently rewrites its behaviour on everything it already knew. In both cases the amount forgotten can only be *measured*, after the fact, on a test set one happens to have kept.

This system takes a third route. It is assembled from **frozen modules**, each publishing a finite, machine-checkable statement of which inputs it is willing to answer. New capacity is added only where the current system *declines* to answer, and nothing already in place is modified. Retention therefore stops being a score and becomes a property that can be checked: across twelve benchmarks and three feature substrates, adding capacity changed **no previously given answers**, verified bit for bit, in every run.

0	0.0000	7/7	none
previously given answers changed by an update — verified bit for bit	answers issued outside a checkable specification, across 45 evaluation runs	datasets where accuracy exceeds the best published adaptive-tuning result	raw samples, exemplars or generators retained between updates

WHAT THE SYSTEM PROVIDES

1. Updates are additive, and provably so. An update can extend what the system answers; it cannot alter an answer already given. This is a structural property of the composition, not a regularisation penalty, so it is verified by comparison rather than estimated on a benchmark.

2. Every answer is covered by a checkable statement, and the system may decline. In all 45 evaluation runs the fraction of answers issued outside a satisfied specification was **0.0000**. Every accuracy figure here counts a declined input as an *error*, so none is flattered by abstention.

3. Every structural change is gated on held-out data. Reorganisations that would make the system more accurate or more compact are proposed automatically, then have to pass an acceptance test before they are adopted, and a refusal is recorded with its reason. Over 66 gated decisions on two independent blocks of unseen seeds, 24 changes were adopted, one of them a regression — 0.7 accuracy points, below the test's own resolution. Of 57 consolidations replayed against real held-out data, none showed negative evidence; in a separate audit of 21 historical decisions the test refuses all 4 known-harmful changes.

4. Nothing raw is retained. No exemplar buffer and no generative replay of samples: the published specifications are what the system rehearses against. This matters where the training data cannot be kept — retention periods, personal data, licensed corpora.

ACCURACY AGAINST PUBLISHED STATE OF THE ART

The standard exemplar-free class-incremental protocol for pre-trained models: a frozen public vision encoder (ViT-B/16, ImageNet-21k), classes revealed in equal steps, nothing stored between steps, class order shuffled per seed. Figures are last-step accuracy on the full test set after the final step, averaged over three evaluation seeds never used during development. Percentages.

		published			*this system*	
dataset	protocol	SimpleCIL	APER	RanPAC	accuracy	stable
CIFAR-100	100 classes / 10 steps	81.3	87.5	92.2	**90.6**	86.7
ImageNet-R	200 / 10	54.6	69.5	77.9	**76.3**	69.5
ImageNet-A	200 / 10	48.9	50.0	62.4	**55.9**	51.6
CUB-200	200 / 10	86.7	87.1	90.3	**89.1**	85.4
ObjectNet-200	200 / 10	53.6	58.2	—	**63.9**	53.6
OmniBenchmark	300 / 10	73.2	74.5	79.9	**76.6**	71.1
VTAB	50 / 5	84.4	85.4	92.2	**93.9**	88.2

SimpleCIL and **APER** (best per-dataset variant of the two reported): Zhou et al., *Revisiting Class-Incremental Learning with Pre-Trained Models*, arXiv:2303.07338v1, supplementary tables, ImageNet-21k block of the split matching ours; last-step accuracy. **RanPAC**: McDonnell et al., NeurIPS 2023, Table 1; its metric is the final average of per-step accuracies and its per-dataset encoder checkpoint is not stated in the paper, so that column is close to but not identical in definition to the others. ObjectNet is not evaluated in RanPAC. **accuracy** and **stable** are two configurations of this system, both fully covered by checkable specifications; only *stable* carries the bitwise-additive guarantee (see overleaf).

The accuracy configuration is **above every published APER variant, and above SimpleCIL, on all seven datasets**. Against RanPAC — the strongest published exemplar-free result on this protocol — it is ahead on 1 dataset, within 1.6 points on 3 more, and behind by 3.3 and 6.5 points on the other two. No published method in this comparison reports whether its updates change answers it had already given. The whole table was then re-run on three further seeds that had never been used: no accuracy cell moved by more than 1.4 points, and the additive guarantee held on all 21 runs.

On the smaller exemplar-free benchmarks the same picture holds with the comparison that is hardest to argue with: on split-MNIST class-incremental the stable configuration reaches **89.3%** storing nothing, against 87.1% for replay trained on exactly the same data while retaining 100 raw images per class.

WHAT IT COSTS, STATED PLAINLY

Accuracy and bitwise stability are two configurations, not a slider. The stable configuration is additive by construction. The accuracy configuration re-derives one readout when a step completes, which moves 7 to 105 of every 100 previously given answers over a run. We tried the obvious remedy — freezing that readout the way the modules are frozen — and it does deliver exact stability, at a cost of 21 to 54 accuracy points, for a reason we can state precisely and which no amount of engineering inside that design removes. We report the two configurations separately rather than blur them.

The encoder is frozen and public. These results ride on the quality of a pre-trained feature extractor; we do not claim gains for training an encoder from scratch, and the system inherits the encoder's blind spots. An optional one-off adaptation step on the first batch of classes is used above; on one of the seven datasets it slightly *lowers* feature quality, and the internal gate caught that on all three seeds without being told.

The gate needs held-out data, and says how much. The acceptance test is a statistical one: it reports the smallest regression it could have detected, rather than failing silently on a small validation split.

BEYOND IMAGES

The same system, with no code or setting changed, has run on two intent-classification streams over a frozen sentence encoder (77 and 150 classes in 11 and 10 steps) and on an intrusion-detection stream of raw network-flow records where attack categories appear a pair at a time. The guarantees carried over unchanged, and on all three the accuracy configuration matched or exceeded replay trained on the same data (93.9 / 96.5 vs. 91.8 / 95.6 on text, 95.1 vs. 95.1 on the tabular stream). Where the simplest configuration was weak on the tabular records, the system's own gate refused every reorganisation built on it — correctly.

THE AXIS THE PUBLISHED WORK DOES NOT OCCUPY

property	this system	published continual learning
answers changed by an update	**0**, verified bit for bit, every run	measured after the fact, never guaranteed
answers outside a checkable specification	**0.0000** over 45 evaluation runs	not reported
structural changes gated on held-out data	yes; refusals logged with reasons	no gate
raw data retained	**none**	buffer or generator
a module rebuilt from its published specification alone	agrees with the original on 92% of its region	not applicable
the system's own forecast of its future drift	published before each change and met in 49 of 54 events against a 25% budget	not applicable

HOW THESE NUMBERS WERE PRODUCED

This is a research programme run to an evaluation discipline that anyone can audit from the repository:

• Development seeds and evaluation seeds are disjoint. Every threshold and configuration was frozen before an evaluation seed was run; the acceptance test, the one component tuned later, is confirmed a second time on a further untouched block of seeds.

• No number is typed by hand. Every figure in this document, including each one above, is regenerated by a single program from per-seed raw records; the 629 result files behind it are content-hashed.

• Published comparisons come from the source PDFs, with table, metric, encoder and protocol recorded per cell — an audit that has already corrected one of our own anchor rows and retired three of our own claims. Refusals and negative results are reported as results.

WHERE THE DETAILS ARE

How a specification is represented and solved, how declining is induced in an already-frozen module, how the acceptance test is calibrated, and the parameter budgets are in the repository that produced this overview and reproduces every row of it: `APPROACH.md`, `TECHNICAL.md`, and `REPORT.md` with every negative result.

WHERE IT FITS

The value is not a point on a leaderboard — it is an update you can defend: regulated classification, clinical and industrial inspection, financial-crime typologies that grow monthly, edge deployments that cannot phone home to retrain.

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